

BEC-351 Fundamentals of AI/ML: From the Turing Test to World Models

Lecture 1 — A Short History, and What It Teaches Us

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Today, in five acts

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GEORGE HARRISON
ALL THINGS MUST PASS



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- ▶ Learning from experience
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Naming trivia

McCarthy chose the name “Artificial Intelligence” partly to keep the new field at arm’s length from *cybernetics* — and from having Norbert Wiener as its guru. Fields, like startups, are sometimes named for who they want to avoid.

Alan Turing — the question before the field (1912–1954)



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Wartime work:

- ▶ Enigma code-breaking at Bletchley Park
- ▶ Estimated (by some historians) to have shortened WWII by years

The birth certificate — Dartmouth, summer 1956

The proposal's authors:

- ▶ John McCarthy — coined “Artificial Intelligence”; later created Lisp
- ▶ Marvin Minsky — co-founder of the MIT AI Lab
- ▶ Claude Shannon — father of information theory
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“Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.”

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Outcome

A 6–8 week brainstorming summer founded a research field — and its first hype cycle: confident predictions of human-level machines “within 20 years.”

Early wins (1955–1966): three programs, three lessons

Logic Theorist (1956)

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- ▶ Proved theorems from
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- ▶ Lesson: reasoning can be
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ELIZA (1966)

- ▶ Weizenbaum's chatbot "psychotherapist"
- ▶ Simple pattern matching — yet users confided in it
- ▶ Lesson: humans *project* understanding. Keep this in mind for every LLM debate you will ever have.

The First AI Winter (1974–1980)

What went wrong:

- ▶ Overpromising, underdelivering
- ▶ Combinatorial explosion — toy demos didn't scale
- ▶ Common-sense reasoning: nowhere in sight
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What it cost:

- ▶ DARPA cut AI funding
- ▶ British support collapsed post-Lighthill
- ▶ Academic positions disappeared

The lesson

Hype is a loan. Winters are how the field pays it back — with interest.

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Genuine successes:

- ▶ **DENDRAL (1965–):** chemical structure analysis — the prototype
- ▶ **MYCIN (1970s):** medical diagnosis; matched specialists in trials
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Why they worked: narrow, well-defined domains; explicit rules; clear business value.

Remember this pattern — *narrow and well-specified* is where AI keeps succeeding first, in every era.

The Second AI Winter (1987–1993)

The bubble burst because expert systems were:

- ▶ Expensive to build *and* maintain
- ▶ Brittle — no graceful handling of uncertainty
- ▶ Bottlenecked on knowledge acquisition: rules had to be typed in by hand

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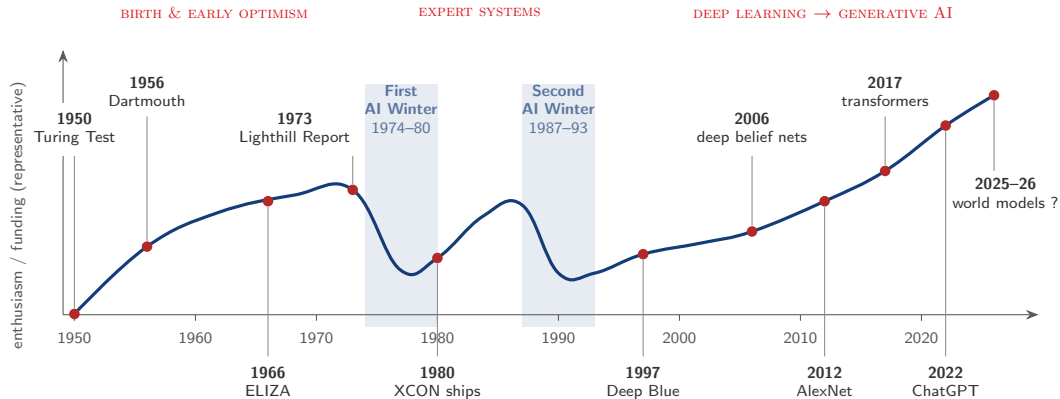
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Silver lining

The field emerged humbler, more mathematical, and more empirical — the culture that would make machine *learning* respectable. (Notice what failed: typing knowledge in. Notice what would win: *learning it from data*.)

Seventy years in one curve



The curve is schematic with correct dates

What thawed the winters: three exponential gifts

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- ▶ Distributed computing
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None of these were built for AI. AI was the beneficiary.

GPU vs CPU: why the accident mattered

CPU:

- ▶ Few powerful cores
- ▶ Sequential processing
- ▶ Complex instruction sets
- ▶ Great for general computing

GPU:

- ▶ Thousands of simple cores
- ▶ Massively parallel processing
- ▶ Built for matrix operations
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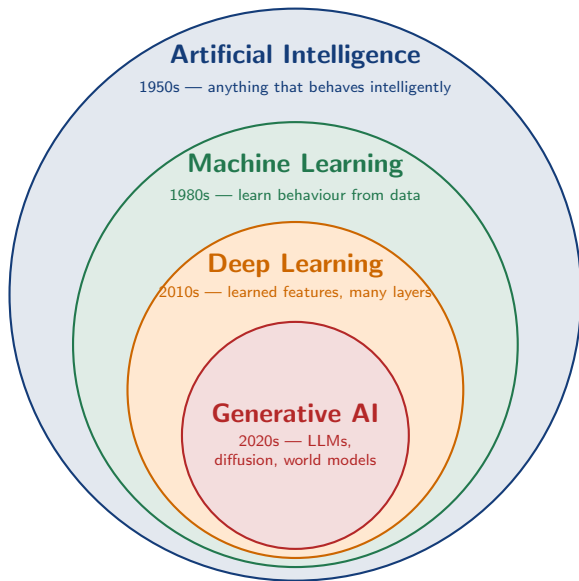
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Key insight — and a course preview

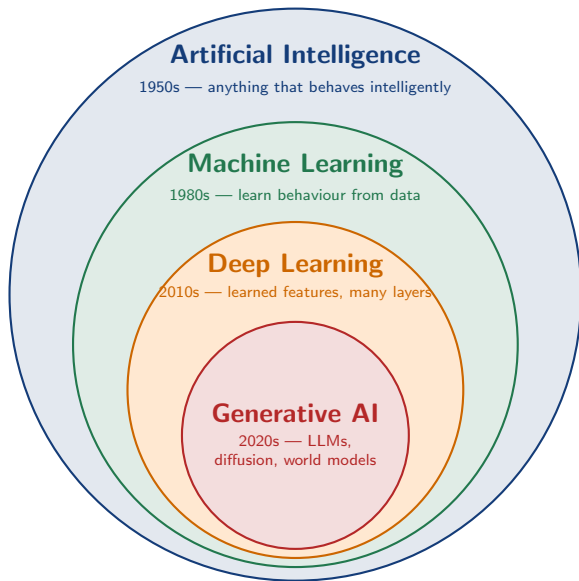
AI workloads are *embarrassingly parallel*: mostly enormous matrix–vector products. When we say in the Linear Algebra review that “a model is matrices,” this slide is the hardware receipt.

AI $\supset$ ML $\supset$ DL $\supset$ Generative AI



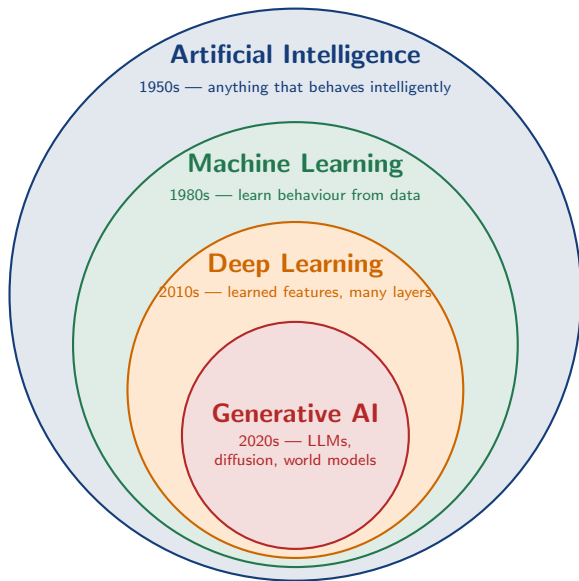
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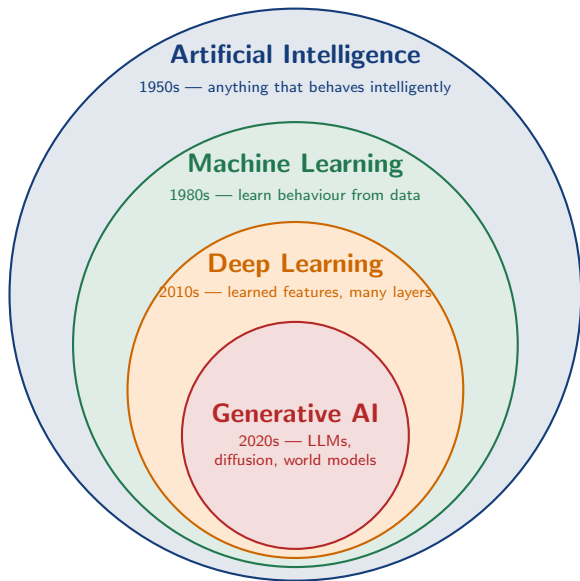
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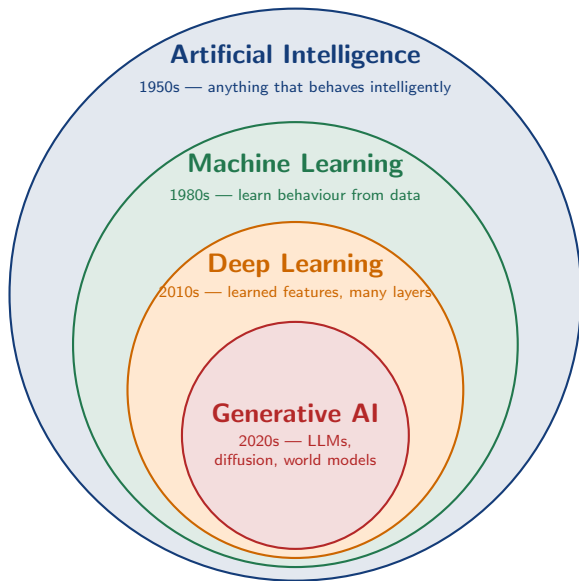
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Each ring keeps the outer ring's goal and automates one more layer of human effort. This course lives mostly in rings 2–3 — deliberately: that is where the transferable mathematics is.

The Deep Learning Revolution (2006–2012)

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The three enablers (memorise this triple):

- ▶ **Data** (ImageNet's 1.2M labelled images)
- ▶ **Compute** (GPUs)
- ▶ **Algorithms** (ReLU, dropout, better initialisation)

2017 → 2026: from Attention to World Models

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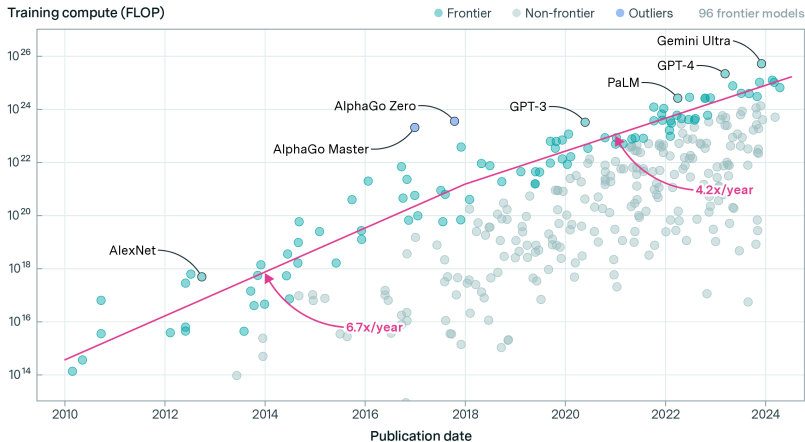
The pattern to notice

Every jump on this slide is the same three ingredients as 2012 — data, compute, algorithms — at ever larger scale.

The compute curve behind it all

Training compute of frontier models

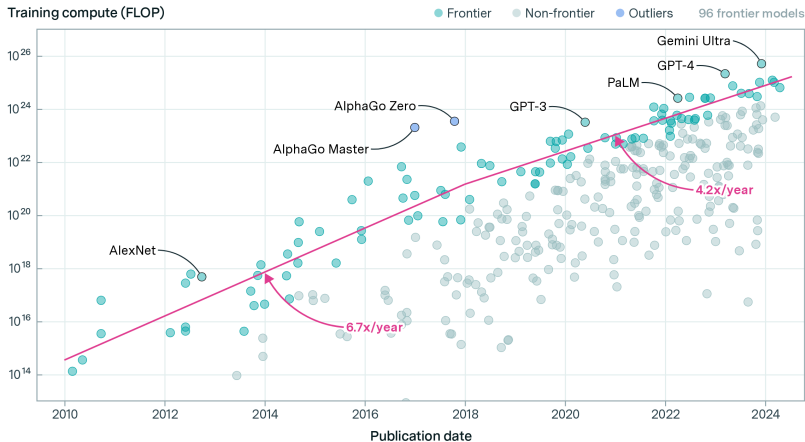
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Notice the year graph is ending!

The three (and a half) types of machine learning

	Supervised	Unsupervised / Self-supervised	Reinforcement
Learns from	Labelled data	Unlabelled data (labels manufactured from the data itself)	Interaction with an environment
Goal	Predict labels	Find structure; learn representations	Maximise long-run reward
Examples	Spam detection Image classification	Clustering, PCA LLM pretraining	Game playing Robotics RLHF fine-tuning

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Where do LLMs sit?

Astride the table: *self-supervised* pretraining (predict the next token — the labels are free) + *reinforcement* fine-tuning (RLHF). Modern systems are pipelines of types, not single cells of this table.

AI today — capabilities and honest limitations

Genuinely strong:

- ▶ Pattern recognition
- ▶ Translation; code generation
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The right column is, not coincidentally, a syllabus: generalisation theory, energy-based modelling, validation, Bayesian uncertainty... we will meet each one.

Future directions and challenges

Technical:

- ▶ Artificial General Intelligence?
- ▶ Explainable AI
- ▶ Data- and energy-efficient learning
- ▶ Multimodal + embodied understanding

Societal:

- ▶ Jobs and skill displacement
- ▶ Bias and fairness
- ▶ Privacy and surveillance
- ▶ Misinformation at scale

AI for Engineering & Science — where this department intends to live

The moral of 70 years — and the case for this course

What every winter killed:

- ▶ Products, companies, funding
- ▶ Specific architectures and toolchains
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Hence: this course

Frameworks will change possibly before you graduate. The mathematics of this course is the part of the field that has survived every boom and every bust since 1956. That is precisely why we teach *fundamentals*.

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The real question

Can someone from *this* class push the frontier of AI forward?

Some of you might already be closer than you think.

Questions!

Next lecture: the Linear Algebra review begins — bring the course map.